Lab 8: Text Editors (September 21, 2026)

Compression

1. Create a sample 10 MB text file by running this command:
   `for i in {1..440000}; do echo "Hello World Line $i"; done > sample.txt`

   Use gzip, bzip2, xz, and zip to compress `sample.txt` file, and use the command `ls -lh` to compare the sizes (No captured screens needed).

   a. Size of gzip file = ...1.2M.....
   b. Size of bzip2 file = ...483K.....
   c. Size of xz file = ...159K.....
   d. Size of zip file = ...1.2M.....

2. TAR command
   a. Using `tar`, backup all files and subdirectories under your home directory. Place the resulting tarball file in the `/tmp` directory, giving it the name `backup.tar`.

```
root@ubuntu:~$ tar cvf backup.tar /tmp
tar: Removing leading `/' from member names
/tmp/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-logind.service-4PmPb6/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-logind.service-4PmPb6/tmp/
/tmp/.X11-unix/
/tmp/.XIM-unix/
/tmp/github-remote/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-timesyncd.service-n19dWJ/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-timesyncd.service-n19dWJ/tmp/
/tmp/.ICE-unix/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-polkit.service-fZRSF0/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-polkit.service-fZRSF0/tmp/
/tmp/theia_upload/
/tmp/theia-install-run
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-ModemManager.service-BZ1eLx/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-ModemManager.service-BZ1eLx/tmp/
/tmp/http-remote/
/tmp/.font-unix/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-resolved.service-oFo4Eb/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-resolved.service-oFo4Eb/tmp/
root@ubuntu:~$ ls -lh
total 20K
-rw-r--r-- 1 root root 20K Sep 21 09:38 backup.tar
lrwxrwxrwx 1 root root   1 Sep 19 15:06 filesystem -> /
root@ubuntu:~$
```

   b. Accomplish the same task with `gzip` compression using the `-z` option with tar, creating `/tmp/backup.tar.gz`.

```
root@ubuntu:~$ tar zcvf backup.tar.gz /tmp
tar: Removing leading `/' from member names
/tmp/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-syste
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-syste
/tmp/.X11-unix/
/tmp/.XIM-unix/
/tmp/github-remote/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-syste
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-syste
/tmp/.ICE-unix/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-polki
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-polki
/tmp/theia_upload/
/tmp/theia-install-run
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-Moder
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-Moder
/tmp/http-remote/
/tmp/.font-unix/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-syste
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-syste
root@ubuntu:~$ ls
backup.tar  backup.tar.gz  filesystem
root@ubuntu:~$ ls -lh
total 24K
-rw-r--r-- 1 root root 20K Sep 21 09:38 backup.tar
-rw-r--r-- 1 root root 602 Sep 21 09:39 backup.tar.gz
lrwxrwxrwx 1 root root   1 Sep 19 15:06 filesystem -> /
```

c. Make backups using the `-j` option with `bzip2` compression, and `-J` option with `xz` compression.

```
root@ubuntu:~$ tar Jcvf backup.tar.xz /tmp
tar: Removing leading `/' from member names
/tmp/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-logind.service-4PmPb6/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-logind.service-4PmPb6/tmp/
/tmp/.X11-unix/
/tmp/.XIM-unix/
/tmp/github-remote/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-timesyncd.service-n19dWJ/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-timesyncd.service-n19dWJ/tmp/
/tmp/.ICE-unix/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-polkit.service-fZRSF0/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-polkit.service-fZRSF0/tmp/
/tmp/theia_upload/
/tmp/theia-install-run
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-ModemManager.service-BZ1eLx/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-ModemManager.service-BZ1eLx/tmp/
/tmp/http-remote/
/tmp/.font-unix/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-resolved.service-oFo4Eb/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-resolved.service-oFo4Eb/tmp/
```

```
root@ubuntu:~$ tar jcvf backup.tar.bz /tmp
tar: Removing leading `/' from member names
/tmp/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-logind.service-4PmPb6/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-logind.service-4PmPb6/tmp/
/tmp/.X11-unix/
/tmp/.XIM-unix/
/tmp/github-remote/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-timesyncd.service-n19dWJ/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-timesyncd.service-n19dWJ/tmp/
/tmp/.ICE-unix/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-polkit.service-fZRSF0/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-polkit.service-fZRSF0/tmp/
/tmp/theia_upload/
/tmp/theia-install-run
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-ModemManager.service-BZ1eLx/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-ModemManager.service-BZ1eLx/tmp/
/tmp/http-remote/
/tmp/.font-unix/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-resolved.service-oFo4Eb/
/tmp/systemd-private-6dce0b7a39214b51906ba4a9c6318632-systemd-resolved.service-oFo4Eb/tmp/
```

d. Compare the size of all four backup files.

```
root@ubuntu:~$ ls -lh
total 48K
-rw-r--r-- 1 root root 20K Sep 21 09:38 backup.tar
-rw-r--r-- 1 root root 605 Sep 21 09:41 backup.tar.bz
-rw-r--r-- 1 root root 20K Sep 21 09:41 backup.tar.gz
-rw-r--r-- 1 root root 588 Sep 21 09:41 backup.tar.xz
lrwxrwxrwx 1 root root   1 Sep 19 15:06 filesystem -> /
```

Nano

3. Use nano to create a file named nano1.txt containing the following text

11111
22222
33333
44444
55555

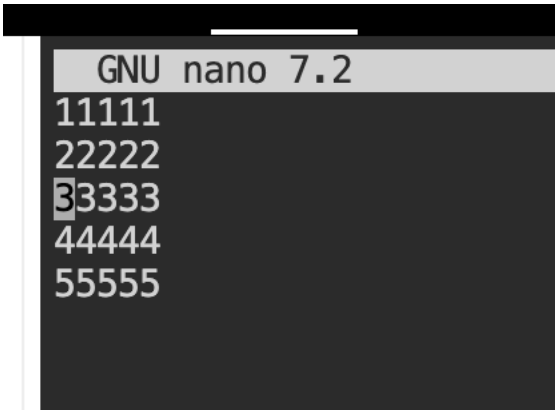
```
root@ubuntu:~$ cat nano1.txt
11111
22222
33333
44444
55555
root@ubuntu:~$
```

4. Use cat command to create a file named nano2.txt containing the following text

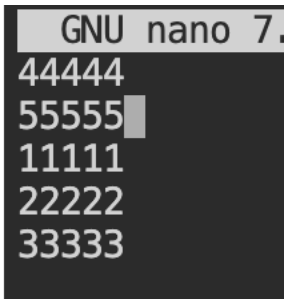
aaaaa
bbbbb
ccccc

```
root@ubuntu:~$ cat nano2.txt
aaaaa
bbbbb
ccccc
root@ubuntu:~$
```

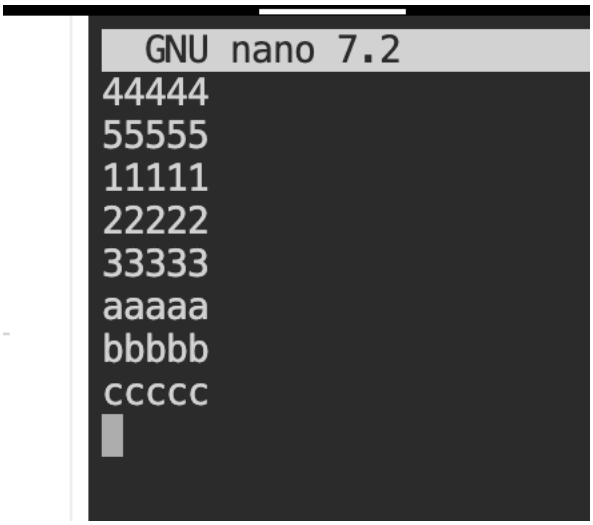
5. Search “333” in nano1.txt (`^w`) Note: you can skip this for Windows guys!



6. In nano1.txt, cut last two lines and paste to the beginning of the file (`^K` for cut and `^U` for paste)



7. In nano1.txt, import content of nano2.txt into the end of the file. (^R)



Vi Lab

8. Use vi to create a file named vifile.txt containing the following text

Vim is a free and open-source, screen-based text editor program.
It is an improved clone of Bill Joy's vi. Test Text Hello Class.
Vim's author, Bram Moolenaar, derived Vim from a port of the Stevie editor for Amiga and released a version to the public in 1991.
Vim is designed for use both from a command-line interface and as a standalone application in a graphical user interface.
Since its release for the Amiga, cross-platform development has made it available on many other systems.
Test1 Test2 Test3
In 2018, it was voted the most popular editor amongst Linux Journal readers;
in 2015 the Stack Overflow developer survey found it to be the third most popular text editor, and in 2019 the fifth most popular development environment.
Vim is released under the Vim license, which includes some charityware clauses that encourage users who enjoy the software to consider donating to children in Uganda.

The Vim license is compatible with the GNU General Public License through a special clause allowing the distribution of modified copies under the GNU GPL version 2.0 or later.

Vim is highly customizable and extensible, making it an attractive tool for users who demand a large amount of control and flexibility over their text editing environment.

XXXXXXXXXX Text input is facilitated by a variety of features designed to increase keyboard efficiency.

Users can execute complex commands with "key bindings," which can be customized and extended.

Test4 Test5 Test6

Test7 Test8 Test9

The "recording" feature allows for the creation of macros to automate sequences of keystrokes and call internal or user-defined functions and mappings.

Abbreviations, similar to macros and key mappings, facilitate the expansion of short strings of text into longer ones and can also be used to correct mistakes.

Vim also features an "easy" mode for users looking for a simpler text editing solution.

```
root@ubuntu:~$ vi
root@ubuntu:~$ cat vifile.txt
Vim is a free and open-source, screen-based text editor program.
It is an improved clone of Bill Joy's vi. Test Text Hello Class.
Vim's author, Bram Moolenaar, derived Vim from a port of the Stevie editor for Amiga and released a version to the public in 1991.
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Test1 Test2 Test3
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Users can execute complex commands with "key bindings," which can be customized and extended.
Test4 Test5 Test6
Test7 Test8 Test9
The "recording" feature allows for the creation of macros to automate sequences of keystrokes and call internal or user-defined functions and mappings.
Abbreviations, similar to macros and key mappings, facilitate the expansion of short strings of text into longer ones and can also be used to correct mistakes.
Vim also features an "easy" mode for users looking for a simpler text editing solution.
```

9. Show line number (`:set number`)

```
  1 Vim is a free and open-source, screen-based text editor program.
  2 It is an improved clone of Bill Joy's vi. Test Text Hello Class.
  3 Vim's author, Bram Moolenaar, derived Vim from a port of the Stevie editor for Amiga and released a version to the public in 1991.
  4 Vim is designed for use both from a command-line interface and as a standalone application in a graphical user interface.
  5 Since its release for the Amiga, cross-platform development has made it available on many other systems.
  6 Test1 Test2 Test3
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 12 XXXXXXXXXX Text input is facilitated by a variety of features designed to increase keyboard efficiency.
 13 Users can execute complex commands with "key bindings," which can be customized and extended.
 14 Test4 Test5 Test6
 15 Test7 Test8 Test9
 16 The "recording" feature allows for the creation of macros to automate sequences of keystrokes and call internal or user-defined functions and mappings.
 17 Abbreviations, similar to macros and key mappings, facilitate the expansion of short strings of text into longer ones and can also be used to correct mistake
 18 Vim also features an "easy" mode for users looking for a simpler text editing solution.
 19
~
~
~
~
~
```

10. Move cursor to
    a. First line (`:0`)

    b. Last line (`:$`)

c. Line 5 (`:5`)

d. Move to end of line 10 (`$`)

e. Move to the first word of line 10 (`^`)

f. Move to upper position of monitor (`H`)
g. Move to in the middle of monitor (`M`)
h. Move to lower position of monitor (`L`)

11. Deletion
    a. Delete the first word in the line 12 (`dw`)
    b. Delete last two worlds in the last line by using command `x`
    c. Delete line 6 (`dd`)
    d. Delete line 13 to line 14 (`:13,14d` or move to line 13 then `2dd`)
    e. Go to the second line, and delete everything from word "Test" to the last word in this line (`D`)
12. Searching
    a. Search the word "Vim" for forward and backward positions.
    b. Count how many "Vim" words in this file. (`:%s/Vim//gn`)
        i. 2matches 2 on 2 lines
13. Replace "is" with "was" everywhere.
14. Copy lines 3 - 4 and paste at the last line.
15. Undo the previous operation (`u`)
16. Switch between line 1 and line 2.
17. Capture the screen and paste it below. Save the file and exit.


Editor
Tab 1
+
35m
1 It was an improved clone of Bill Joy's vim.
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14 Abbreviations, similar to macros and key mappings, facilitate the expansion of short strings of text into longer ones and can also be used to correct mwastakes.
15 vim also features an "easy" mode for users looking for a simpler text.
16
1 line less; before #47 2 seconds ago